CST1510 Programming for data communication and networks

Week 1: Getting Started with Python

LAB ACTIVITY

Welcome to your first Python lab session! This week, we'll focus on Python's foundational building blocks, including variables, basic data types (int, float, string), and essential input/output operations.

By the end of this session, you'll be well-equipped to move forward in your Python journey.

Topics Covered:

- **Variables and data types (int, float, string)**
- **Basic input (input()) and output (print())**
- **Introductory Python syntax**

This week is all about becoming comfortable with the core elements of Python. These fundamental concepts will be the backbone for everything you'll do in this module. The more you practice, the more natural it will feel.

During this lab, you'll complete several exercises designed to reinforce the core concepts of Python. You aim to work through as many activities as possible to deepen your understanding.

Once you've completed the tasks, select one or two activity demonstrating what you learned this week to include in your portfolio.

Remember, your portfolio should reflect your comprehension of variables, data types, and basic input/output operations—so choose wisely!

Problem-Solving & Pseudocode Guide

1. Understand the Problem:
    - Identify the **inputs**, expected **outputs**, and any **constraints**.
2. Break Down the Problem:
    - Divide the problem into smaller steps. Each step should focus on a single task, like getting input, performing calculations, or displaying results.
3. Write Pseudocode:

- Describe each step in simple language, focusing on the logic rather than code syntax.

In [7]:
```
#Example:
#1. Get the width and height from the user
#2. Calculate area (width * height)
#3. Calculate perimeter (2 * (width + height))
#4. Display area and perimeter
```

4. Convert to Code:
   - Translate your pseudocode into actual programming code, step-by-step.

In [8]:
```
# Step 1: Get width and height from user
width = float(input("Enter the width of the rectangle: "))
height = float(input("Enter the height of the rectangle: "))

# Step 2: Calculate area
area = width * height

# Step 3: Calculate perimeter
perimeter = 2 * (width + height)

# Step 4: Display area and perimeter
print(f"The area of the rectangle is: {area}")
print(f"The perimeter of the rectangle is: {perimeter}")
```

```
The area of the rectangle is: 8.0
The perimeter of the rectangle is: 12.0
```

ACTIVITY:

Threshold:

1. Questioner

Create a program asking few questions and printing the answer.
The questions could be as simple as: Name, interest, nationality, course they are studying on, experience in programming, etc..

Your program should read your name from the user and the output should contains some escape sequence character

In [1]:
```
# Instructions
# 1. We can ask a question by using the user input function
# example input('Whats your name)
# 2. The input could be stored in a variable called 'name' so we can use
# 3. Now we can create a print statement in which we could print a bit of
# incapsulated in quotation marks and pass variable 'name' containing the
# The above two lines of code could be repeated for different questions
# We still can use user input for questions and the print statement for o
```

```
# The variable name will change
# CREATE TWO QUESTIONS AND ANSWERS
```

```
The history saving thread hit an unexpected error (OperationalError('attem
pt to write a readonly database')).History will not be written to the data
base.
```

**2. Area and perimeter of a rectangle. **

Write a program that displays the area and perimeter of a rectangle with a width of 4.5 and a height of 7.9 using the following formulas:

area=width x height
perimeter=2* (width+height)

In [9]:
```
# Instructions: When you need to display somthings for sure we need to us
#1. create a variable 'width' and assign value 4.5 in it

#2. create a variable 'height' and assign value 7.9 in it

#3. craete a variable area and assign the variables width * height in it
# (which will automatically get the value from the variables width and mu

#4. create a variable perimeter which will hold this formula 2 * (width +

#5. Print the area and perimeter to the screen and add some text to know
```

3. Convert pounds into kilograms

Write a program that converts pounds into kilograms. The program prompts the user to enter a pound value, converts it to kilograms, and displays the result. One pound is 0.454 kilograms. Here is a sample run:

Enter a value in pounds:55.5 5.55 pounds is 25.197 kilograms

In [10]:
```
#1. Create variable pounds that store and convert the user input to a num

#2. Create variable kg storing the formula for converting from lbs to kg.

#3. Print the result on the screen.
```

4. Convert feet into meters

Write a program that reads a number in feet, converts it into meters, and displays the result. One foot is 0.305 meters. Here is a sample run:

Enter a value for feet:16.5 16.5 feet is 5.0325 meters

In [11]:
```
#1. Create variable feet that stores and converts the user input to a num

#2. Create variable kg storing the formula for converting from lbs to kg.

#3. Print the result on screen.
```

5. Sum of the First n Positive Integers

Write a program that reads a positive integer, n, from the user and then displays the sum of all of the integers from 1 to n. The sum of the first n positive integers can be computer using the formula:

$$sum = \frac{n*(n+1)}{2}$$

In [12]:
```
# Get the number from the user and converting it to a numerical value, in

# Calculating the sum of all the numbers from 0 to n by using the formula

# Printing the sum on screen. (use the f formating in your output)
```

6. Geometry: area of a pentagon

Write a program that prompts the user to enter the side of a pentagon and displays the area. Format output into two decimal places.

The area of a pentagon can be computed using the following formula (s is the length of a side):

$$area = \frac{5*s^2}{4*tan(\frac{\pi}{5})}$$

In [13]:
```
# Importing some mathematical functions from the math module/library

# Gettig the side length of the pentagon.

# Calculating and saving the area of the pentagon in the area variable

# Printing the result on the screen
```

Typical

7. Time of watching Netflix series

Write a program to calculate how long it would take you to watch nonstop Netflix series that comprises of: 5 Seasons, 10 Episodes each, of which episodes in 1st season are 45 min and 35 sec each, and the remaining ones are 35 min and 15 sec each.

8 Distance between Two Points on Earth The surface of the Earth is curved, and the distance between degrees of longitude varies with latitude. As a result, finding the distance between two point on the surface of the Earth is more complicated than simply using the Pythagorean theorem. Let the (t_1,g_1) and (t_2,g_2) be the latitude and longitude of two points on the Earth's surface. The distance between these points following the surface of the Earth, in kilometres is:

Distance = 6371.01 x arccos(sin(t_1) x sin(t_2) + cos(t_1) x cos(t_2)x cos(g_(1)- g_(2)))

The value 6371.01 in the previous equation wasn't selected at random. It is the average radius of the Earth in kilometres.

Create a program that allows the user to enter the latitude and latitude and longitude of two points on the Earth in degrees. Your program should display the distance between the points, following the earth's surface, in kilometres.

Hint: Python's trigonometric functions operate in radians. As a result, you will need to convert the user's input from degrees to radians before computing the distance with the formula discussed previously. The math module contains a function named radians which converts from degrees to radians.

9. Write a programme calculating the amount of money you will have in 3 years, knowing that you will put aside £10 every week. During the first year, your parents will be giving you £20 every week. Your little sister will be taking out of your savings 0.50 every 2 weeks.

Excellent

10. (Population projection) The US Census Bureau projects population based on the following assumptions:

One birth every 7 seconds One death every 13 seconds One new immigrant every 45 seconds

Write a program to display the population for each of the next five years.

Assume the current population is 312032486 and one year has 365 days.

Hint: in Python, you can use integer division operator // to perform division.

The result is an integer. For example, 5 // 4 is 1 (not 1.25) and 10 // 4 is 2 (not 2.5).

11. Write a program that reads the following information and prints a payroll statement: Employee's name (e.g., Smith) Number of hours worked in a week (e.g., 10) Hourly pay rate (e.g., 9.75) Federal tax withholding rate (e.g., 20%) State tax withholding rate (e.g., 9%)

A sample run is shown below:

```
Enter employee's name: Smith
Enter number of hours worked in a week: 10
Enter hourly pay rate: 9.75
Enter federal tax withholding rate: 0.20
Enter state tax withholding rate: 0.09

Employee Name: Smith
```

Payroll statement:

```
Hours Worked: 10.0
Pay Rate: $9.75
Gross Pay: $97.5
Deductions:
  Federal Withholding (20.0%): $19.5
  State Withholding (9.0%): $8.77
  Total Deduction: $28.27
Net Pay: $69.22
```

In [ ]: